

Open-Source Reference for Reproducible Side-Channel Assessment of Post-Quantum Standards

Dillibabu Shanmugam, Patrick Schaumont

Worcester Polytechnic Institute, USA {dshanmugam, pschaumont}@wpi.edu

Abstract—We present the first truly open-source post-quantum cryptography silicon reference: a unified RISC-V coprocessor that runs ML-KEM-1024 and ML-DSA-44 in unmasked and first-order DOM-masked configurations on a fully open FPGA flow. The 38-instruction CV-X-IF coprocessor sits on a CV32E40X RV32IMC SoC. The flow uses Yosys, nextpnr-xilinx, and Project X-Ray to target the CW305 board. All configurations pass the FIPS-203/204 KAT tests. Our objective is to achieve *reproducible* power side-channel leakage assessment in support of the ongoing transition to post-quantum standards, which must be completed within the next few years. Indeed, the open-source nature of the platform and flow is important because the side-channel leakage assessment methodology for PQC is not yet standardized or established. There is a need for a widely shared platform to support reproducible experiments. Furthermore, power side-channels are substantially harder to model than timing side-channels, especially if low-level, sub-cycle hardware effects must be taken into account. Our open reference design demonstrates both protected and unprotected PQC implementations. On masked ML-KEM-1024, a CCA-PC TVLA cuts first-order leakage by $\geq 23\times$ (masked 3.10σ at $N=10k$ vs. naive 71.55σ at $N=5k$), with pre-silicon (RTeAAL) and post-silicon (CW305) results in agreement.¹

Index Terms—Post-quantum cryptography, power side-channel evaluation, open-source hardware, RISC-V coprocessor.

I. THE CASE FOR OPEN-SOURCE IN PQC POWER SCA

While NIST finalized standards for post-quantum cryptographic key exchange (ML-KEM-1024) and signatures (ML-DSA-44) in August 2024, the timeline for deployment is extremely tight: NIST IR 8547 recommends transition by 2030; CNSA 2.0 calls for 2035. Standard PQC provides algorithmic protection against quantum-computing-based attacks, but its *implementations* must also be validated against classic timing-based and power-based side-channels. In recent years, several open-source verification tools for timing side-channels have been demonstrated [2], [3]. However, the verification of power side-channels, essential for implementations that operate at the edge, is not yet available. Timing effects are measured in clock cycles, but power effects extend into the sub-cycle and circuit-level domain. For example, glitches as well as power-domain coupling have been identified as root causes of power side-channel leakage [4]. Therefore, the standard practice to verify power side-channels is to build and measure a prototype implementation. In the case of PQC, such prototypes include PQC software stacks, PQC hardware coprocessors, and PQC custom ISA extensions.

The challenge of reproducibility. A prototype for power side-channel verification of PQC is complex. It covers the source code, the tool flow, the target architecture, the measurement setup, and the post-processing methodology for measured traces. For emerging PQC standards, this reproducibility is a must, because the assessment methodology *itself* is not yet standard, and because the PQC standards focus on functionality and do not specify or guide the secure implementation of the standard. Every abstraction level in the implementation modulates the power side-channel [9]:

TABLE I
COVERAGE OF PRIOR OPEN-SOURCE PQC HARDWARE AND SCA TOOLING.

Work	KEM/SIG	Masking	Target	Silicon TVLA
HORCRUX [8]	both	—	FPGA	—
HOPE-MLKEM [11]	KEM	high-order	FPGA	—
OpenTitan/OTBN	FW	—	FPGA	—
This work	both	1st-order DOM	FPGA	Saarinen-PQC

memory layout, register allocation, forwarding, netlist, routing. A PQC SCA result can be reproduced between labs only if every layer is documented and visible. Achieving reproducibility in this context is challenging; even state-of-the-art venues such as CHES typically demonstrate the assessment on standard (black-box) components such as micro-controllers. To support PQC roll-out, we propose a fully open-source implementation and tool-flow, and we make sure that the complete experiment is portable and reproducible.

The challenge of PQC side-channel assessment. While the PQC standards have been selected, the proper side-channel assessment testing is still being debated. Existing methodologies for classical crypto such as AES build on three ingredients: a fixed-vs-random class definition that names the leakage target, a sampling discipline, and a positive control. ISO/IEC 17825:2024 contains no PQC-specific test class. Saarinen’s HOST work-in-progress [5] proposes a PQC adaptation describing several such test cases (for example, ML-KEM Decaps targets sk , m , K' , and the PRF seed, each defining a different experiment). Yet, agreeing on the assessment methodology in the context of *open-source* artifacts is a must. For example, NDAs forbid testing labs from publishing pass/fail decisions on proprietary silicon and commercial parts. Measurement on a public reference therefore remains the only practical path, and an open PQC platform is the precondition for refining both the test criteria and the countermeasures those criteria will eventually verify.

Related open-source PQC and SCA tooling. Table I positions this work against prior open-source PQC hardware and pre-silicon SCA tooling.

Contributions. (i) A unified 38-instruction CV-X-IF coprocessor running both ML-KEM-1024 and ML-DSA-44 on one RTL base, with first-order masking on every secret-touching kernel. (ii) An open FPGA flow (Yosys + nextpnr-xilinx + Project X-Ray) onto CW305, with all configurations passing the FIPS-203/204 KAT byte-exact on silicon, and an open ASIC reference flow (Yosys + OpenROAD + KLayout, IHP-Open-PDK SG13G2). (iii) An ISO 17825 compliant test of the protected implementation of the proposed ML-KEM.

II. THREAT MODEL AND SCOPE

Our threat model assumes the attacker has physical proximity, ChipWhisperer-class power capture, and chosen-input access. Invasive probing, fault injection, EM, and higher-order DPA are not considered in this experiment, but are available to testers of our open-source reference. The design must resist first-order DPA

¹Release: <https://github.com/Secure-Embedded-Systems/open-source-reference-for-reproducible-side-channel-assessment-of-Post-Quantum-Standards>

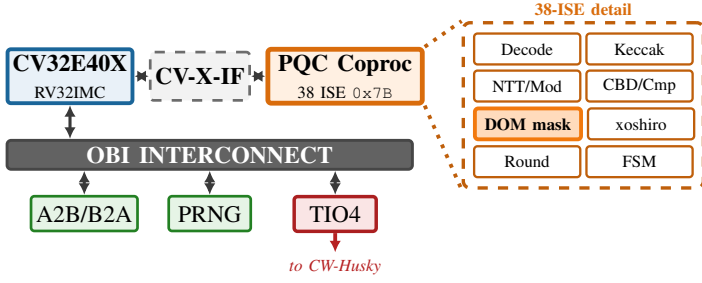


Fig. 1. PQC coprocessor SoC: CV32E40X over CV-X-IF, with A2B/B2A, PRNG, and TIO4 trigger on OBI. Dashed box: 8 ISE groups, DOM-mask highlighted.

TABLE II
ML-KEM-1024 DECAP: SW vs. ISE vs. ISE+DOM (CW305, CV32E40X @ 16 MHz, LIVE `mcycle` READ).

Configuration	Decap (k cyc)	Speedup	LUT (Δ)
SW-only (this work)	7,402	1.0 $\times$	28,445 (–)
ISE	180	41 $\times$	32,618 (+14.7%)
ISE + DOM	401	18 $\times$	32,618 (+0%)

on every secret-touching kernel and avoid mask-share collapse under synthesis; a post-synthesis share-cone separation pass ships with the release.

III. OPEN-SOURCE ARCHITECTURE AND BACKEND FLOW

Figure 1 shows the SoC. A CV32E40X RV32IMC core attaches the 38-instruction PQC coprocessor over CV-X-IF on Custom-3 opcode 0x7B. Eight pipelined units cover Decode, Keccak, NTT/Mod, CBD/Cmp, DOM mask, xoshiro128++ PRNG, Rounding, and Trap/FSM. First-order DOM masking sits inside the secret-touching gadgets; linear kernels operate share-wise. The same SystemVerilog source feeds Yosys and forks into two backends: FPGA (nextpnr-xilinx, Project X-Ray, CW305 XC7A35T at 16 MHz) and ASIC (OpenROAD, KLayout, IHP SG13G2 130 nm), with no vendor IP in either path. For pre-silicon cross-validation of the FPGA path we adopt the RTeAAL Sim tensor-algebra RTL simulator [10] and compare its per-cycle toggle traces against the post-silicon CW305 captures.

IV. SILICON EXPERIMENT

A. KAT closure on CW305

The DUT is a CW305 with XC7A35T-2 at 16 MHz; cycles are read live from the RISC-V `mcycle` CSR.

Three-way comparison. Table II compares the three ML-KEM-1024 implementation styles on the same hardware, with the SW-only Decap cycle count read live on CW305 from the unprivileged `mcycle` CSR (KeyGen 5,892 k, Encap 6,026 k, Decap 7,402 k – so on the same CV32E40X the unmasked SW Decap is dominant). Against this measured baseline, the 38-instruction CV-X-IF ISE delivers a $\sim 41\times$ **speed-up** on Decap. First-order DOM masking costs $2.23\times$ in cycles relative to naive ISE but **zero additional fabric area**: the masked datapath is baked into the ISE RTL and is unused by the unmasked firmware, so the same bitstream serves both via a firmware-only swap. The ISE footprint adds +14.7% LUT, +10.2% FF, and +48 DSP48E1 over the SW-only baseline. ML-DSA-44 follows the same pattern (380 k $\rightarrow$ 820 k cycles when masked).

B. First-order TVLA on masked ML-KEM-1024 decapsulation

Our evaluation protocol follows Saarinen’s proposed PQC adaptation of ISO 17825 [5], inheriting the $N \geq 10k$ pool size and 4.5σ Welch- t threshold from the symmetric-cipher framework and targeting the leakage class at the Fujisaki–Okamoto share-recombine window. At that window the implicit-rejection comparison $c \stackrel{?}{=} c'$ unmarks m' , and the plaintext-check oracle bit

TABLE III
SHARE-RECOMBINE TVLA ON ML-KEM-1024-1024 DECAP (SAARINEN-PQC, 4.5σ WELCH- t THRESHOLD). POST-TRIGGER STATISTICS; ALL THREE CONFIGS SHARE THE SAME BITSTREAM RTL, SAME TRIGGER WINDOW, SAME A/B POOL, CAPTURED AT 16 MHz ON CW305.

Configuration	N/pool	1st-ord. $ t $	2nd-ord. $ t $	Var ratio $ 1-r $	Verdict
SW (no ISE issued)	10k	13.96σ	3.17σ	0.0701	LEAK
ISE naive (pos. control)	5k	71.55σ	53.82σ	0.85	LEAK
ISE + DOM (masked)	10k	3.10σ	4.01σ	0.0784	PASS

transiently exists. This is exactly where the published PC-oracle attacks [6], [7] extract the secret. Our two fixed-vs-random pools draw ciphertexts from distinct seed PRNGs gated to the same cycle and Hamming-weight envelope, isolating that bit.

Table III reports the result on three configurations captured under the same Saarinen-PQC protocol on the same CW305 board. The masked ISE design holds at $|t|=3.10\sigma$ ($N=10k$) against a 71.55σ unmasked-ISE positive control ($N=5k$, saturated early) on byte-identical firmware – a $\geq 23\times$ suppression. A pure-software ML-KEM-1024 reference (no Custom-3 instructions issued; ISE present in fabric but unused) captured on the same bitstream-RTL combination at the ISO-17825 floor of $N=10k$ shows 13.96σ , a real first-order leak but $\sim 5\times$ weaker than the ISE-naive baseline projected to the same N ($71.55\sigma \cdot \sqrt{2} \approx 101\sigma$). The dilution is consistent with the SW path running the share-recombine operation across 7 402 k cycles on the CV32E40X core versus 180 k cycles on the dedicated ISE datapath – same secret event, smaller per-cycle power density.

Interpretation. The masked design clears the 4.5σ threshold at first order against both unmasked baselines. The 2nd-order $|t|=4.01\sigma$ also clears, by 0.49σ , below the positive spike expected of an alive $d=1$ DOM gadget at this N . Aliveness rests on the byte-identical FIPS-203 KAT pass, source-level review of the masking gadget, and the public capture pipeline. A positive 2nd-order control needs $N \geq 10^5$ per pool and is future work. The SW row’s first-order leak is broadband – 149 out of 5 750 post-trigger samples exceed 4.5σ but the largest run of consecutive crossings is shorter than the ISO 17825 default of five samples – which is the signature expected of a secret-dependent operation diluted across millions of cycles rather than concentrated in a narrow window.

V. PRE-SILICON EVALUATION

Pre-silicon evaluation identifies vulnerable components before tape-out and steers the next-iteration fix. While a silicon TVLA returns only a single pass/fail number, the same fixed-vs-random pool replayed at two views of the circuit pinpoints the responsible register or wire. We adopt RTeAAL-based tensor simulation [10] at the RTL netlist (L1, per-bit) and the routed XC7A35T fabric annotated with open chipdb PIP-timing context (L3, per-tile and per-PIP). The shared pool indexing lets a leak identified at L1 back-trace deterministically through `synth_xilinx` techmap to specific PIPs, and an L3 tile flagged as leaky back-attributes to the originating RTL register.

For our masked ML-KEM-1024 (Table IV), the trace points to the three unmasked CV-X-IF boundary registers (rs1, rs2, result) as the source of leakage, not the DOM gadget. The masked core stays clean at both layers, and the silicon CW305 power measurement at the same N agrees. The fix follows directly: re-mask the boundary registers before they cross the CV-X-IF latch, at a cost of one extra PRNG cycle and one XOR stage of latency. Closed-source vendor TVLA flows cannot produce per-component maps of this kind. Our simulator tracks signal activity but not gate-level glitches; parasitic coupling, glitch propagation, and PVT-corner timing are left to a follow-up gate-level simulation.

TABLE IV
CROSS-LAYER LEAKAGE ATTRIBUTION AT $N=10K$.

Region	Mask	L1 bits	L3 PIPs (sig.)	L3 tiles $\cap$ reg.
Unmasked CV-X-IF boundary	✗	96/96	14,951	248
Masked core	✓	0/240	58/2,947	713/2,593

VI. CONCLUSION

We present a fully open-source reference of the new PQC standards, with and without countermeasures, and demonstrate a successful assessment of the first-order leakage of the design.

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